Year 10 Mathematics

Sets 1

9.3 Three-Set Problems

Three-Set Problems

STARTER · 5 MIN

Date: _____ **Title:** Three-Set Problems

LEARNING OBJECTIVES

- Complete a Venn diagram for three sets
- Find unions and intersections of three sets
- Use numbers of elements in three-set diagrams

STARTER *Answer in your book. No calculators.*

1. List the multiples of 2 up to 12
2. List the multiples of 3 up to 12
3. List the factors of 12
4. Which number is in all three lists?

Starter — Answers

STARTER

Tick or correct in green pen

1. **2, 4, 6, 8, 10, 12**

2. **3, 6, 9, 12**

3. **1, 2, 3, 4, 6, 12**

4. **6 and 12**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can complete a Venn diagram for three sets	☐
SILVER	I can find unions and intersections of three sets	☐
GOLD	I can use numbers of elements in three-set diagrams	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Three-Set Venn Diagrams

INTRODUCTION · 3 MIN

Three circles make 8 regions, including the region outside all circles.

Centre first

$A \cap B \cap C$ is the middle region.

In all three sets

Then pairs

Fill each overlap of two circles.

$A \cap B$ but not C

Then singles

Fill the parts in one circle only.

Only A

Outside

Elements in none of the sets.

$(A \cup B \cup C)'$

Examples 1 and 2 — Reading a Three-Set Diagram

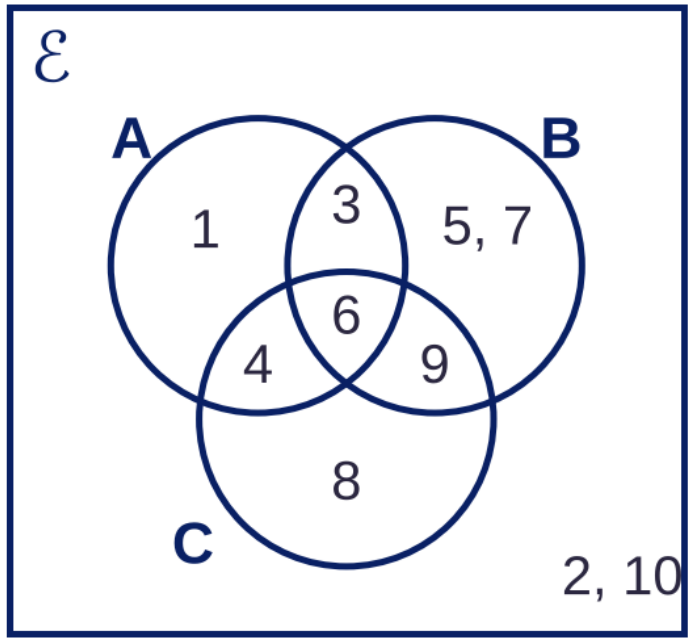
EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

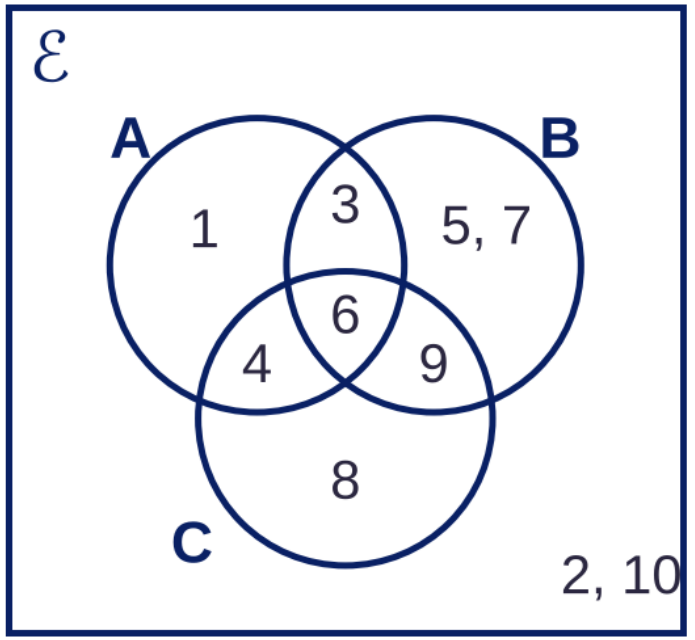
Example 1

List $A \cap B \cap C$.



Example 2

List the elements of A.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

The centre region

$A \cap B \cap C$ **= {6}**

Tip: the middle is in all three circles.

Example 2

All of circle A

A = {1, 3, 4, 6}

Tip: include every part of circle A.

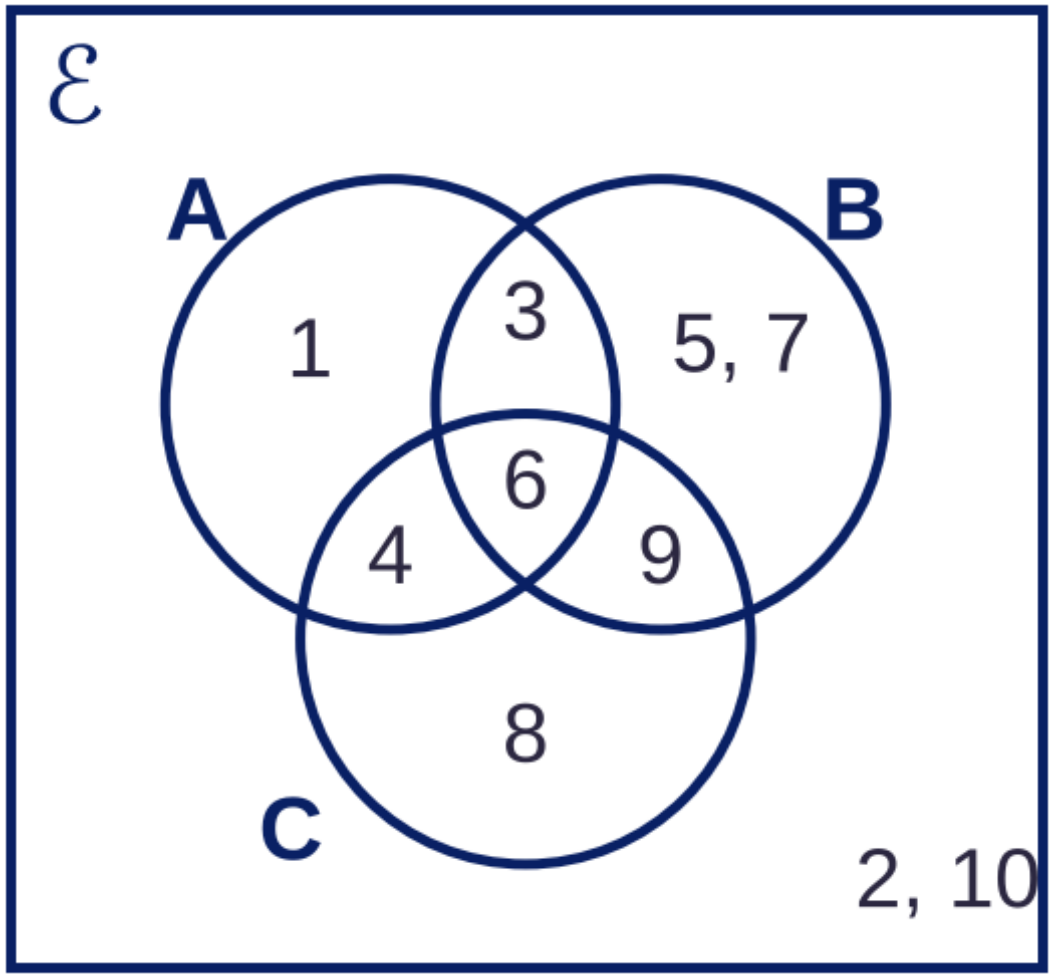
Task 1

PRACTICE · 9 MIN

BRONZE *Use the Venn diagram*

1. List B
2. List C
3. List $A \cap B$
4. List $B \cap C$
5. List $(A \cup B \cup C)'$

★ List $A \cup C$



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $B = \{3, 5, 6, 7, 9\}$
2. $C = \{4, 6, 8, 9\}$
3. $A \cap B = \{3, 6\}$
4. $B \cap C = \{6, 9\}$
5. $(A \cup B \cup C)' = \{2, 10\}$

★ $A \cup C = \{1, 3, 4, 6, 8, 9\}$

Examples 3 and 4 — Completing a Three-Set Diagram

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

$\mathscr{E} = \{1,\ 2,\ \ldots,\ 12\}$

A = {even},
B = {multiples of 3},
C = {factors of 12}

Find A ∩ B ∩ C.

Example 4

Using the sets in Example 3,

find A ∩ C but not B.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 2, 3, 4, 6, 12\}$

$A \cap B \cap C$ **$= \{6, 12\}$**

Tip: find the centre first.

Example 4

$A \cap C = \{2, 4, 6, 12\}$

Remove elements of B: 6 and 12

Answer **$= \{2, 4\}$**

Tip: this is one region of the diagram.

Task 2

PRACTICE · 9 MIN

SILVER

$\mathcal{E} = \{1, 2, \ldots, 12\}$, $A = \{\text{even}\}$, $B = \{\text{multiples of 3}\}$, $C = \{\text{factors of 12}\}$

1. List $B \cap C$
2. List the elements only in A
3. List $(A \cup B \cup C)'$
4. List $A \cup B$
5. Find $n(A \cap B)$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $B \cap C = \{3,\ 6,\ 12\}$

2. Only $A = \{8,\ 10\}$

3. $(A \cup B \cup C)' = \{5,\ 7,\ 11\}$

4. $A \cup B = \{2,\ 3,\ 4,\ 6,\ 8,\ 9,\ 10,\ 12\}$

5. $A \cap B = \{6,\ 12\}$, $n(A \cap B) = 2$

Examples 5 and 6 — Numbers of Elements

EXAMPLES · 5 MIN

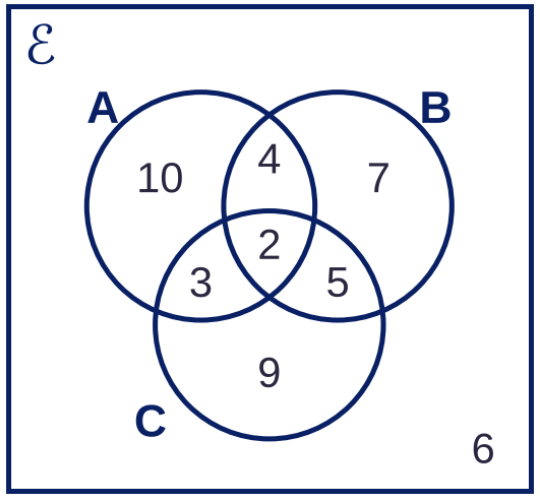
GOLD

Exam-style: show all your working

Example 5

(2 marks)

The diagram shows numbers of elements. Find n(A).



Example 6

(2 marks)

Using the same diagram, find $n(A \cup B)$.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$n(A) = 10 + 4 + 3 + 2$

$n(A) = \mathbf{19}$

Tip: add every region in circle A.

Example 6

$10 + 4 + 7 + 3 + 2 + 5$

$n(A \cup B) = \mathbf{31}$

Tip: count each region only once.

Task 3

PRACTICE · 9 MIN

GOLD

The diagram shows the number of elements in each region

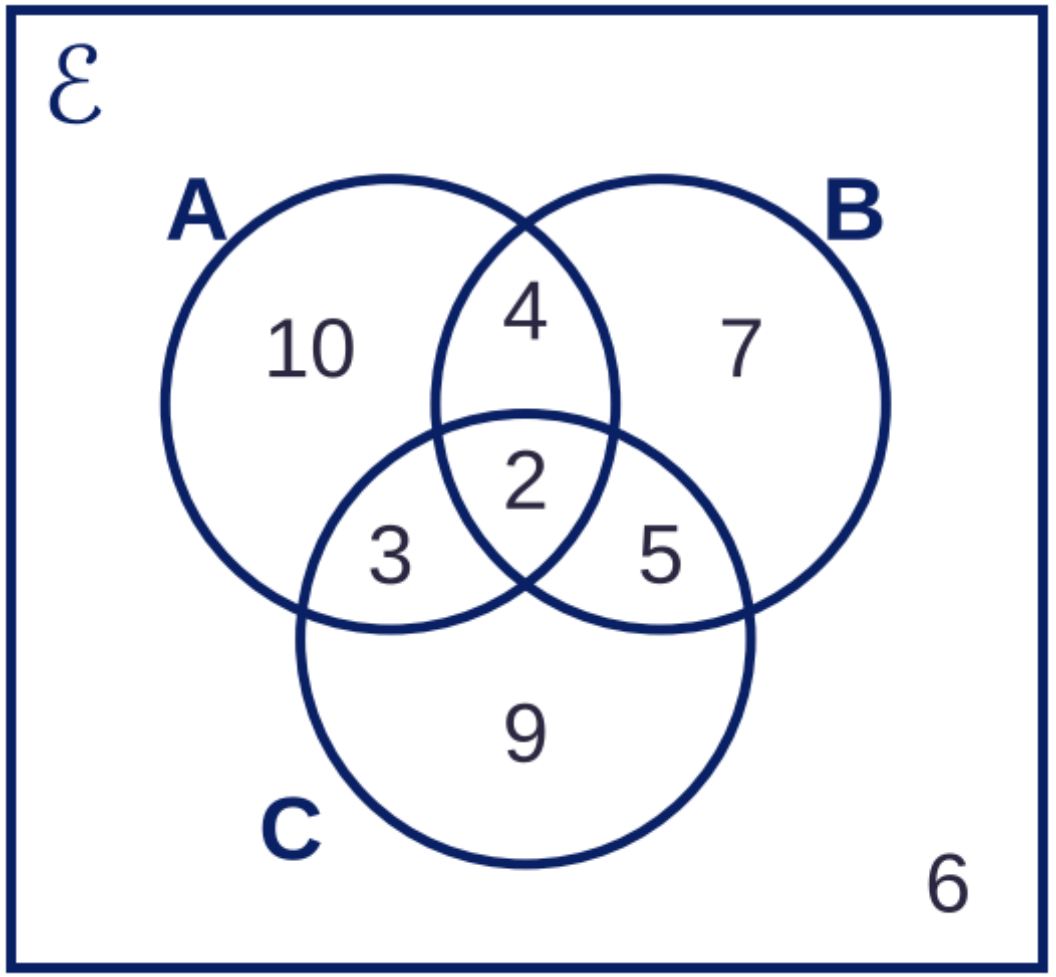
1. Find n(C) (1 mark)

2. Find n(A ∩ B) (1 mark)

3. Find n((B ∩ C)′) (2 marks)

4. Find n(ℰ) (1 mark)

5. Find n(A ∩ B ∩ C′) (2 marks)



Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $n(C) = 9 + 3 + 5 + 2 = \mathbf{19}$

2. $n(A \cap B) = 4 + 2 = \mathbf{6}$

3. $46 - 7 = \mathbf{39}$

4. $10 + 7 + 9 + 4 + 3 + 5 + 2 + 6 = \mathbf{46}$

5. $n(A \cap B \cap C') = \mathbf{4}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE In the Bronze diagram, list $A \cap C$.

SILVER In the Bronze diagram, list the elements only in B.

GOLD In the Gold diagram, find n(B).

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $A \cap C = \{4,\ 6\}$

SILVER Only $B = \{5,\ 7\}$

GOLD $n(B) = 7 + 4 + 5 + 2 = \mathbf{18}$

Exam Practice — Reading a Three-Set Diagram

EXAM PRACTICE · 10 MIN

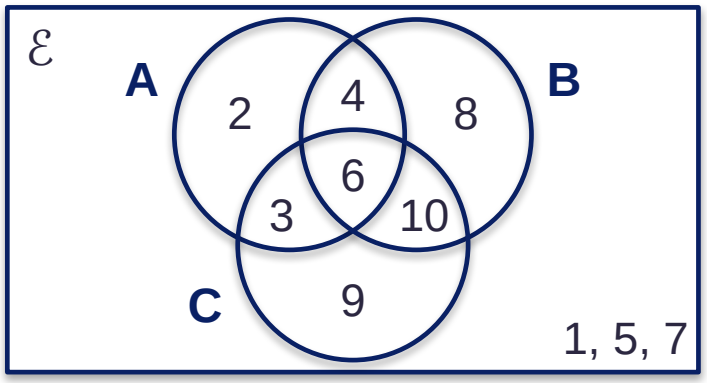
BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

Here is a Venn diagram.



(a) List $A \cap B \cap C$. (1)

.....

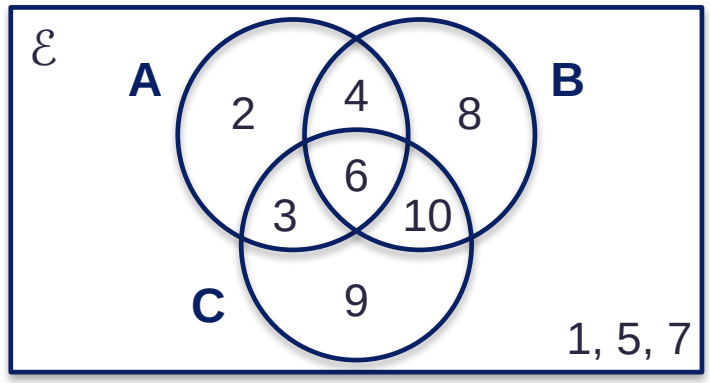
(b) List the elements of B. (1)

.....

Question 2

(2 marks)

Here is a Venn diagram.



(a) List the elements of $A \cap C$. (1)

.....

(b) List $(A \cup B \cup C)'$. (1)

.....

Exam Practice — Completing a Three-Set Diagram

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

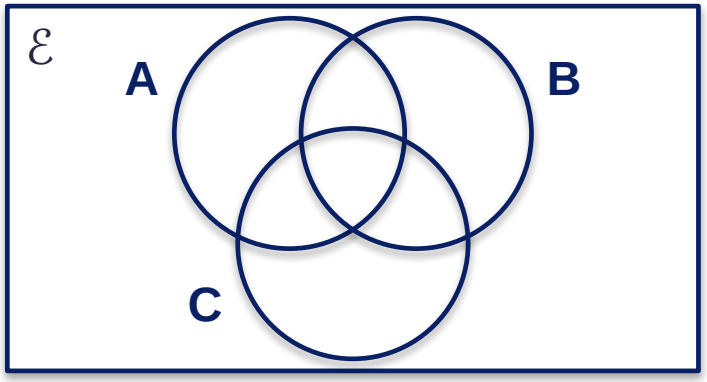
(3 marks)

$\mathcal{E}$ = {1, 2, 3, …, 10}

A = {even numbers}

B = {multiples of 3}

C = {factors of 10}



Question 4

(3 marks)

Use the sets in Question 3.

(a) List the elements of A ∪ B. (1)

.....

(b) List the elements of B ∩ C. (1)

.....

(c) Find n(A ∩ C). (1)

n(A ∩ C) = ...

Exam Practice — Numbers of Elements

EXAM PRACTICE · 10 MIN

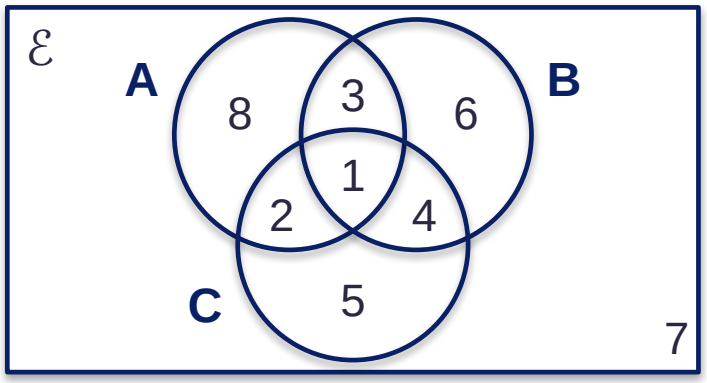
GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

Numbers are counts of elements.



(a) Find n(A ∩ B). (1)

n(A ∩ B) = ...

(b) Find n((A ∪ C)′). (2)

.....

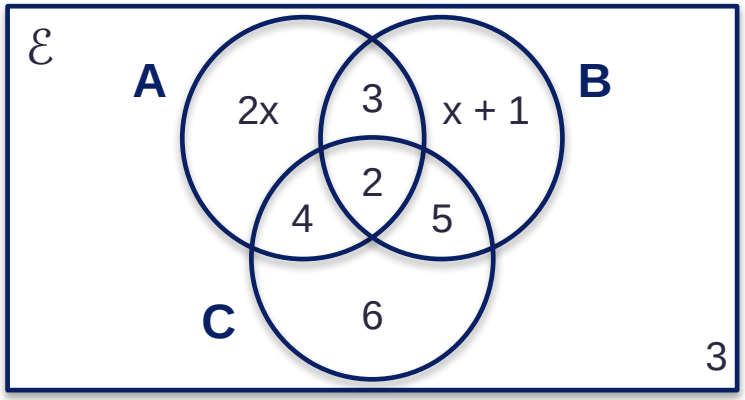
Question 6

(3 marks)

n(ℰ) = 30

Numbers are counts of elements.

Find n(B).



.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) {6} — B1

(b) {4, 6, 8, 10} — B1

Tip: the centre is in all three circles.

Question 2

(a) {3, 6} — B1

(b) {1, 5, 7} — B1

Tip: A ∩ C includes the centre region.

Exam Practice — Mark Scheme

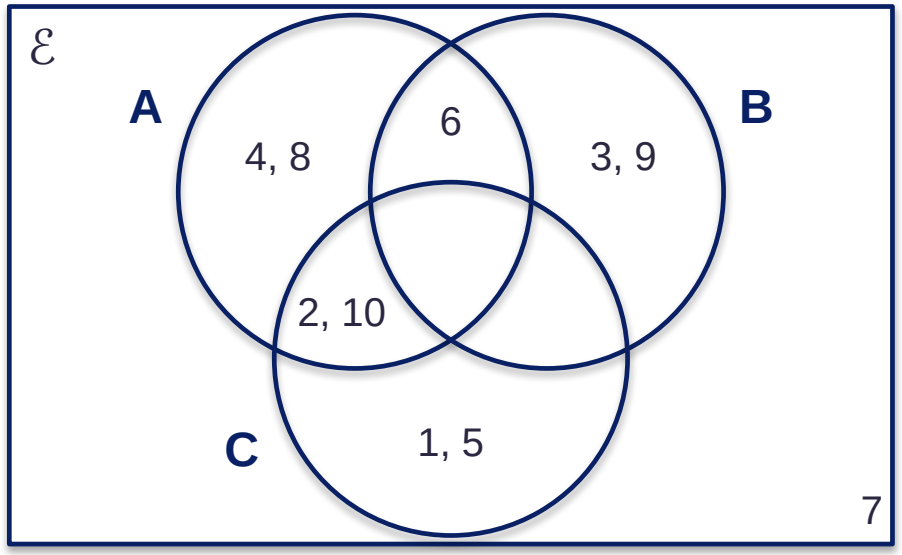
CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

All 10 numbers correct B3



Tip: B2 for 8 or 9 correct, B1 for 6 or 7.

Question 4

(a) {2, 3, 4, 6, 8, 9, 10} B1

(b) { } or ø B1

(c) {2, 10}, n(A ∩ C) = 2 B1

Tip: ø means no element is in both sets.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) 3 + 1 = 4 — B1

(b) only B and outside — M1

6 + 7 = 13 — A1

Tip: (A ∪ C)′ is everything outside A and C.

Question 6

2x + x + 1 + 23 = 30 — M1

x = 2 — A1

n(B) = 3 + 3 + 2 + 5 = 13 — A1

Tip: all regions add up to n(ℰ).